

# **Fundamentals of AI and ML (CSA2001)**

## **README: AI Travel Planner**

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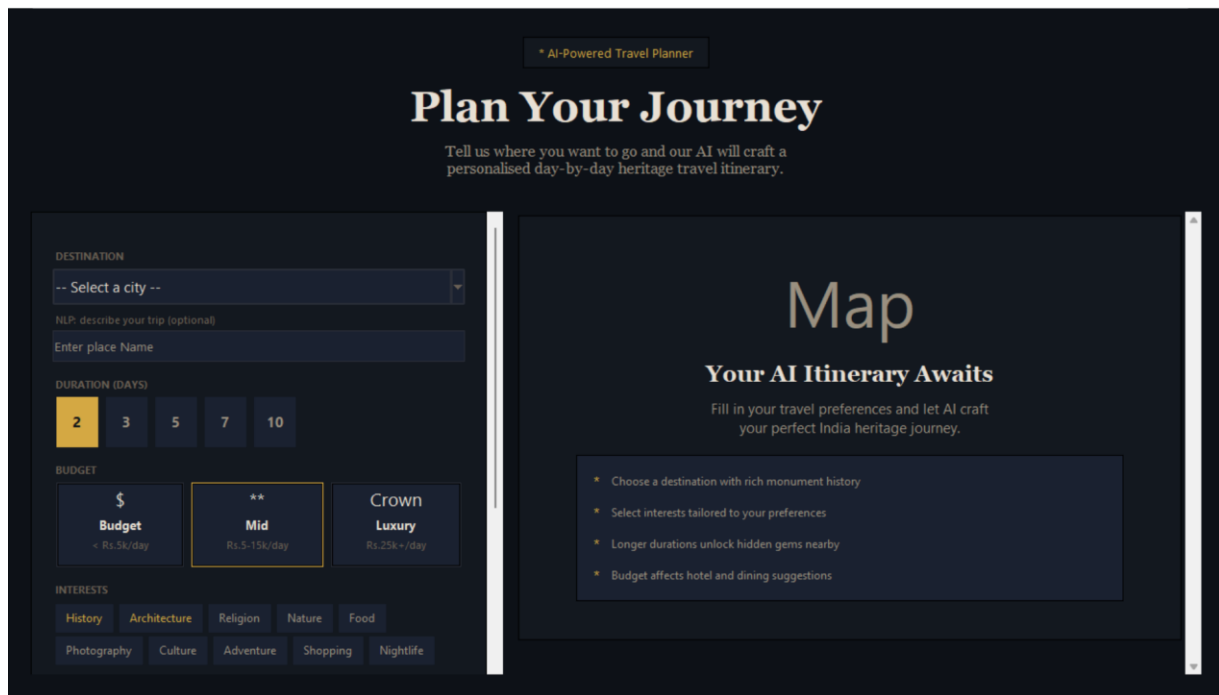
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**Year: 1st Year**

**AI Travel Planner** is a premium Python-based desktop application designed to simplify the creation of heritage-focused travel itineraries. Using a custom-themed interface and a robust PDF engine, it transforms travel data into structured, timeblocked plans.



## Project Overview

This application caters to travellers seeking cultural depth. It features a curated knowledge base of over 15 major Indian heritage hubs, including **Agra, Delhi, Jaipur, Hampi, Varanasi, etc.**

## Key Features

- **Intelligent Scheduling:** Automatically generates detailed itineraries for 2, 3, 5, 7, or 10 days.
- **Time-Blocked Planning:** Daily schedules are divided into **Morning** (7 AM), **Afternoon** (1 PM), and **Evening** (6 PM) slots.
- **Budget Tiering:** Integrated cost estimations for **Budget**, **Mid-range**, and **Luxury** travel styles.

- **Professional PDF Export:** Dynamically creates branded A4 PDF documents using the ReportLab engine.
- **Modern UI:** Features a sleek, dark-themed dashboard with antique gold accents.
- **Simulation Engine:** Provides real-time feedback during processing, such as "Mapping heritage sites by era" and "Optimising travel distances".

DESTINATION

-- Select a city --

NLP: describe your trip (optional)

Enter place Name

DURATION (DAYS)

235710

BUDGET

\$

Budget

< Rs.5k/day

\*\*

Mid

Rs.5-15k/day

Crown

Luxury

Rs.25k+ /day

INTERESTS

HistoryArchitectureReligionNatureFood

PhotographyCultureAdventureShoppingNightlife

TRAVEL STYLE

Walk Through History

Varanasi, Uttar Pradesh Heritage Journey

The Eternal City on the Ganges

3 Days

Mid Budget

Relaxed Pace

3

5

8

Rs.1,000

DAYS

EXPERIENCES

MONUMENTS

EST. DAY

D1

Day 1 - The Ghats & Sacred Fire

Saturday, 04 Apr 2026

7 AM

MORNING - 7 AM

Sunrise Boat Ride on the Ganges

Varanasi's 84 ghats stretch for 6 km along the western bank of the Ganges. A sunrise boat ride (cost - Rs. 300-500) reveals the full spectacle of daily Hindu life — priests, pilgrims bathing, yoga practitioners, and the burning pyres of Manikarnika Ghat. This is considered one of the world's oldest continuously inhabited cities.

Source: Photography Culture

1 PM

AFTERNOON - 1 PM

Kashi Vishwanath Temple

One of the 12 Jyotirlingas (blessed abodes of Lord Shiva) and among the most important temples in Hinduism. The temple has been rebuilt multiple times after destruction; the current golden spire structure was commissioned by Mughal Emperor Akbar in 1595. The corridor was nearly expanded in 2021. Entry: Free.

Temple (Hindu) History

6 PM

EVENING - 6 PM

Ganga Aarti at Dashashwamedh Ghat

The grandest of Varanasi's 84 ghats, Dashashwamedh is where Lord Brahma is said to have performed the Dash Ashwamedh yagna (10 horse sacrifice). Every evening at dusk, a team of priests simultaneously perform a choreographed fire ritual with large brass dhol, conch shells, and burning churning — drawing thousands of devotees and visitors.

Silence (Fire) Sacred

D2

Day 2 - Sarnath & the Old City

Sunday, 05 Apr 2026

7 AM

MORNING - 7 AM

Sarnath — The Deer Park

Located 10km north of Varanasi, Sarnath is one of the four most sacred Buddhist pilgrimage sites. It was here, in the Kashiwan Dham Park, that the Buddha delivered his first sermon (the Dhammacakkapavattana Sutta) in 483 century BCE. The Dhamek Stupa (500 CE) marks the sacred spot. Entry: Rs.25 (devotees).

Buddhist / UNESCO / History

1 PM

AFTERNOON - 1 PM

Ramnagar Fort & Museum

Built in the 18th century by Maharaja Balwant Singh, this cream-coloured sandstone fort across the Ganges is the ancestral seat of the Maharaja of Varanasi. The Veda Vyasa Museum inside houses a remarkable collection of medieval weapons, royal potteries, vintage automobiles, and rare astronomical instruments.

Fort (Mughal) Museum

6 PM

EVENING - 6 PM

Banarasi Silk Weaving Quarter

Varanasi's silk weaving tradition dates back over 2,000 years. The city's weavers — concentrated in the Peshi Kothi and Mandapara neighbourhoods — produce the celebrated Banarasi silk saris known for their gold and silver brocade (zari) work. A guided workshop visit reveals the intricate handloom process. Cost: Free-Rs.200.

Craft (Culture) Shopping

D3

Day 3 - The 84 Ghats — A Complete Walk

Monday, 06 Apr 2026

7 AM

MORNING - 7 AM

Manikarnika Ghat — The Burning Ghat

Manikarnika is one of the most sacred cremation sites in Hinduism — Hindus believe dying here guarantees moksha (liberation from the cycle of rebirth). Fires have burned here continuously for 3,500 years according to tradition. The ghat is managed by the Doot community, hereditary keepers of the cremation fires. Approach with deep respect and no cameras. Entry: Free.

Sacred (Hindu) Culture

1 PM

AFTERNOON - 1 PM

Assi to Raj Ghat — Full Ghat Walk

Walk the 6 km stretch of all 84 ghats from Assi Ghat (south) to Raj Ghat (north) — an extraordinary 3-hour journey through every dimension of Varanasi life. Each ghat has a distinct character: Harischandria (a second burning ghat), Kedar Ghat (South Indian pilgrims), Dashbanga Ghat (a palace), and Panchaganga Ghat (five sacred rivers).

Walking (Culture) Heritage

6 PM

EVENING - 6 PM

Evening Aarti at Multiple Ghats

Beyond Dashashwamedh, smaller aartis happen simultaneously at Assi Ghat (more intimate, popular with locals), Ragendra Prasad Ghat, and Panchaganga Ghat, witnessing the same ancient ritual at three different locations on the same evening gives a sense of the city's layered spiritual geography. Cost: Free.

Ritual (Sacred) Photography

Generated by Walk Through History - 28 Mar 2026, 16:54

## Technical Stack

- **Language:** Python 3.x
- **GUI Framework:** Tkinter (Custom themed)
- **PDF Generation:** ReportLab
- **ML Models:** Nearest Neighbours and Linear Regression from Scikit-Learn.
- **Concurrency:** Threading (ensures the UI remains responsive during document generation)
- **Utilities:** Webbrowser, Datetime, OS, Tempfile

## Installation & Setup

### 1. Clone the Repository: Bash

```
git clone
```

```
https://github.com/Aklesh1206/Tra
```

```
vel-Planner.git cd Travel-Planner
```

### 2. Install Dependencies: Bash

```
pip install reportlab
```

```
pip install scikit-learn
```

```
pip install numpy
```

```
pip install pandas
```

### 3. Run the Application: Bash

```
python Travel.py
```

## How to Use

1. **Select Destination:** Choose a city from the dropdown menu (e.g., Delhi, Mysore, or Mumbai).
2. **Set Preferences:** Choose the trip duration and pace (Relaxed, Packed, Focused, or Flexible).
3. **Generate:** Click "Generate Plan" to begin the AI simulation steps.
4. **View/Export:** Once complete, open the professional PDF or share a digital link to view it in a browser.